

Eenheid 3.7

Note Title

2015/03/26

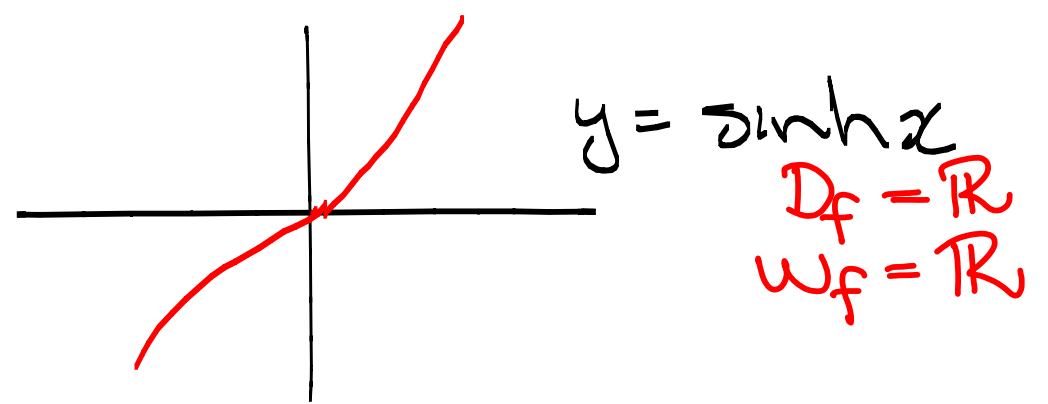
Selfstudie Stewart p 259

Definities van hiperboliese funksies

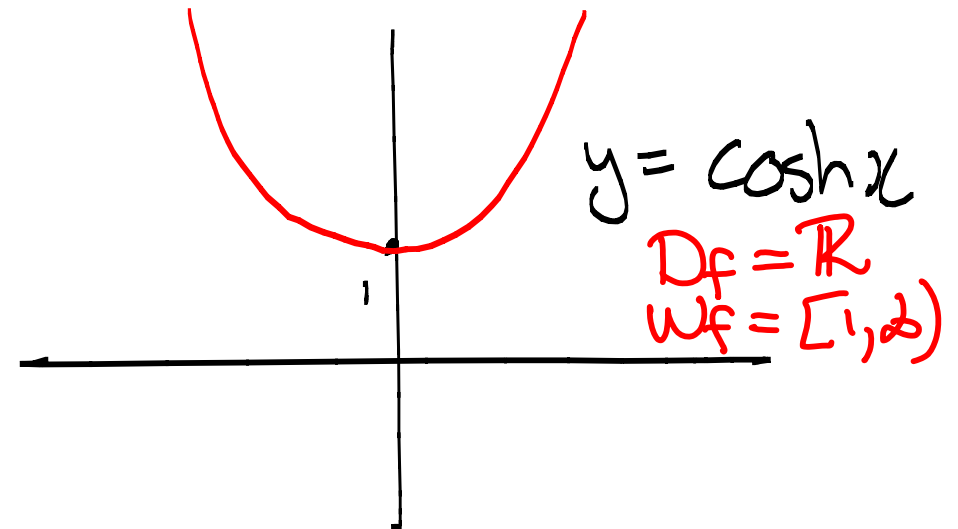
- $y = \sinh x = \frac{e^x - e^{-x}}{2}$
- $y = \cosh x = \frac{e^x + e^{-x}}{2}$
- $y = \tanh x = \frac{\sinh x}{\cosh x} = \frac{e^x - e^{-x}}{e^x + e^{-x}}$

Sketse: p 259

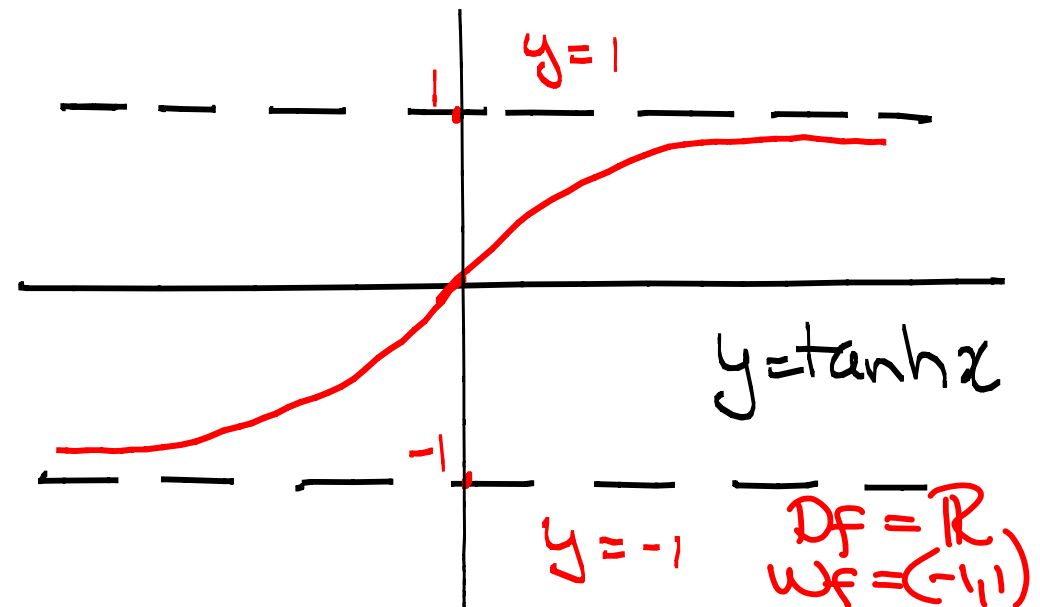
① $y = \sinh x$



② $y = \cosh x$



③ $y = \tanh x$



Aufgeleitet

$$\textcircled{1} \quad \frac{d}{dx} (\sinh x) = \cosh x$$

$$\textcircled{2} \quad \frac{d}{dx} (\cosh x) = \sinh x$$

$$\textcircled{3} \quad \frac{d}{dx} (\tanh x) = \operatorname{sech}^2 x = \frac{1}{\cosh^2 x}$$

Beweis:

$$\begin{aligned} \textcircled{1} \quad \frac{d}{dx} (\sinh x) &= \frac{d}{dx} \left(\frac{1}{2} (e^x - e^{-x}) \right) \\ &= \frac{1}{2} (e^x + e^{-x}) = \cosh x \end{aligned}$$

$$\begin{aligned}
 \textcircled{2} \quad \frac{d}{dx} (\cosh x) &= \frac{d}{dx} \left(\frac{1}{2} (e^x + e^{-x}) \right) \\
 &= \frac{1}{2} (e^x - e^{-x}) \\
 &= \sinh x
 \end{aligned}$$

Voorbeelde

$$\textcircled{1} \quad \frac{d}{dx} \sin(x^4) = \cos x^4 \times 4x^3$$

$$\textcircled{2} \quad \frac{d}{dx} \sinh x^4 = \cosh x^4 \times 4x^3$$

$$\begin{aligned}
 \textcircled{3} \quad \frac{d}{dx} \cosh^3 2x &= \frac{d}{dx} (\cosh 2x)^3 \\
 &= 3(\cosh 2x)^2 \times \sinh 2x \times 2
 \end{aligned}$$

$$\textcircled{4} \quad \frac{d}{dx} (\ln \cosh 4x) = \frac{4 \sinh 4x}{\cosh 4x} \\ = 4 \tanh 4x$$